

Project Report

1. INTRODUCTION

1.1 Project Overview

The project, Automated School IT Ticket Classification and Notification System, is developed using ServiceNow to automate the classification of IT support tickets. The system identifies the issue from the ticket description and automatically assigns the appropriate Category and Subcategory. It also sends an email notification to the Caller.

1.2 Purpose

The purpose of the system is to reduce manual ticket classification, improve data consistency, speed up IT support processes, and provide automated notifications to users.

2. IDEATION PHASE

2.1 Problem Statement

School IT Helpdesk staff receive multiple requests related to Wi-Fi, projectors, passwords, and computer performance. Manually reviewing and categorizing every ticket is time-consuming and may lead to errors. An automated system is required to classify tickets and notify users efficiently.

2.2 Empathy Map Canvas

Section	Content
Think & Feel	Concerned about workload and incorrect ticket classification.
Hear	Frequent complaints about IT issues from students and teachers.
See	Large number of IT support requests requiring manual processing.
Say & Do	Review tickets, identify issues, assign categories, and respond to users.
Pain	Manual classification, repetitive work, delays, and possible errors.
Gain	Faster classification, reduced workload, consistent ticket handling, and timely notifications.

2.3 Brainstorming

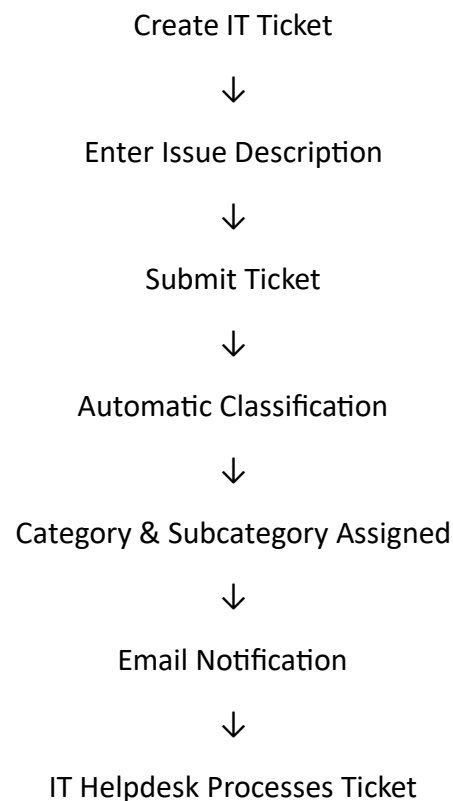
The proposed ideas included:

- Automatic ticket classification.
- Category and Subcategory dependency.

- Flow Designer automation.
- Automated email notification.
- Structured ticket data storage.
- Easy maintenance and future scalability.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

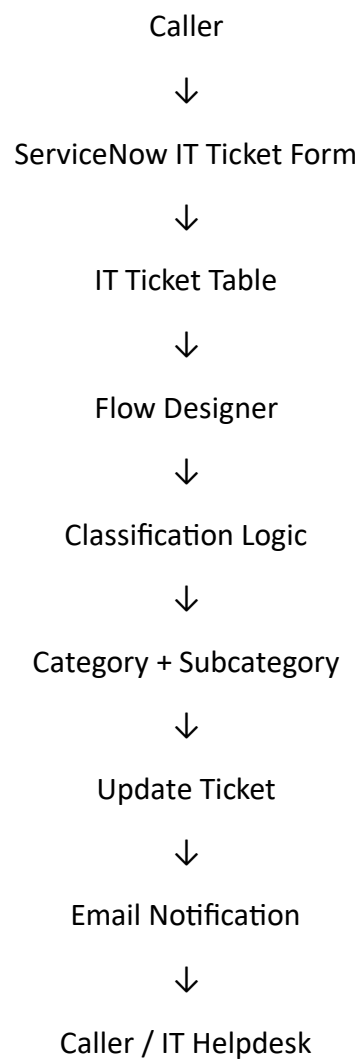


3.2 Solution Requirement

The system must:

- Automatically classify IT tickets.
- Assign Category and Subcategory.
- Support dependent Category–Subcategory choices.
- Send an automated email to the Caller.
- Store ticket information in a structured format.
- Support easy maintenance and future scalability.

3.3 Data Flow Diagram



3.4 Technology Stack

Component	Technology
Platform	ServiceNow
Automation	Flow Designer
Database	ServiceNow Tables
Business Logic	Flow Designer Conditions
Notification	ServiceNow Email
Access Control	ServiceNow Roles and Permissions

4. PROJECT DESIGN

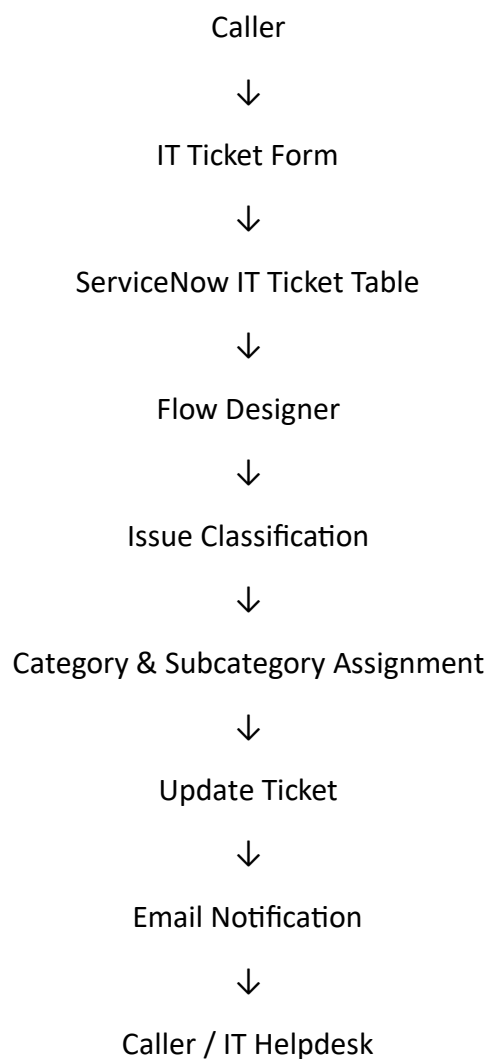
4.1 Problem Solution Fit

The solution directly addresses manual IT ticket classification by automating issue identification and assigning the appropriate Category and Subcategory. It also reduces communication delays through automated email notifications.

4.2 Proposed Solution

The proposed solution uses **ServiceNow Flow Designer** to analyse the Short Description of a newly created ticket. Based on predefined conditions, the system automatically assigns the relevant Category and Subcategory and sends an email notification to the Caller.

4.3 Solution Architecture



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project is divided into **5 Epics and 12 User Stories**, with a total of **31 Story Points**.

Epic	Description	Assigned To
EPIC-1	IT Ticket Table & Field Configuration	Team Members
EPIC-2	Category–Subcategory Dependency	Rohith SD, Ishwarya K
EPIC-3	Automated Ticket Classification	Sakthipriya R, Yuvarani D
EPIC-4	Automated Email Notification	Rohith SD, Ishwarya K
EPIC-5	Testing, Validation & Deployment	Harsha Baskar

Project Start Date: 28/09/2026

Project End Date: 30/09/2026

Team Lead: Harsha Baskar

Team Members: Sakthipriya R, Yuvarani D, Rohith SD, Ishwarya K

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

The system was tested to verify that the automated classification and notification processes work correctly.

Test Case	Input	Expected Output	Result
TC-01	Wi-Fi not working in library	Network → Wi-Fi	Pass
TC-02	Projector not turning on	Hardware → Projector	Pass
TC-03	Wi-Fi ticket	Email sent to Caller	Pass
TC-04	Projector ticket	Email sent to Caller	Pass

The tests confirm that the system automatically classifies the tickets and sends the required email notifications without manual classification.

7. RESULTS:

Output Screenshots:

servicenow

AllFavoritesHistoryWorkspacesAdmin

Update Sets

Search

Update Sets

Name

Your current update set has been changed to Default [Global]

Actions on selected rows...

New

All

	Name	Application	State	Installed from	Created	Created by	Parent	Batch Base
☐	Default	Global	In progress		2026-06-13 02:55:17	system	(empty)	(empty)
☐	Project Update Set	Global	In progress		2026-09-28 06:53:47	admin	(empty)	(empty)

Related Links

[Merge Update Sets](#)

1 to 2 of 2

Incident Workflow [u_incident_workflow]Default view

Form Design

FieldsField Types

Re

Now fields will modify the database structure

Address (Simple)

Currency

Data Structure

Dynamic Attribute Store

FX Currency

IP Address (Validated IPv4, IPv6)

Reference

Incident Workflow [u_incident_workflow]

1 Column

Number

Caller

Category

Subcategory

Short Description

Description

State

Assignment Group

Assigned to

Workflow Studio					
Auto Classify School IT Tickets		Auto Classify School IT Tickets	Auto Classify School IT Tickets	Auto Classify School IT Tickets	Auto Classify School IT Tickets
EXECUTION DETAILS		Auto Classify School IT Tickets		Test Run - Completed	Open flow
Show Action Details		Auto Classify School IT Tickets		Open context record	
FLOW STATISTICS		Run as System Administrator	Open flow logs	Completed	2026-09-28 23:57:41
TRIGGER		Incident WorkFlow Created		Open current record	
ACTIONS					
1 If SD is Wifi or Network		Flow Logic	Evaluated - True	2026-09-28 23:57:41	104ms
2 Update Record		Core Action	Completed	2026-09-28 23:57:41	22ms
3 Else If SD is Projector or Hardware		Flow Logic	Not Run		
4 Update Record		Core Action	Not Run		
5 Else If SD is Forgot password.		Flow Logic	Not Run		
6 Update Record		Core Action	Not Run		
7 Else If SD is Performance		Flow Logic	Not Run		
8 Update Record		Core Action	Not Run		
9 Send Email			Completed	2026-09-28 23:57:41	570ms
ERROR HANDLER					

Workflow Studio

Auto Classify School IT Tickets

Active

Test flow

Run your flow to make sure it has no errors before you activate it. When the test finishes running, check the execution details to see each action's configuration, runtime values, and the log messages for any errors that occurred.

* Incident WorkFlow Record

INC0001004

X

Q

+

ⓘ

Your test has finished running. View the flow execution details.

☐ Run test in background ⓘ

Cancel

Run Test

Data

Collapse All

Flow Variables

Trigger - Record Created

Incident WorkFlow Record

Record

Incident WorkFlow Table

Table

Run Start Time UTC

Date/Time

Run Start Date/Time

Date/Time

1 - If

2 - Update Record

Incident WorkFlow Record

Record

Incident WorkFlow Table

Table

Action Status

Object

3 - Else If

4 - Update Record

Incident WorkFlow Record

Record

Incident WorkFlow Table

Table

Action Status

Object

5 - Else If

6 - Update Record

Incident WorkFlow Record

Record

ERROR HANDLER

ⓘ

If an error occurs in your flow, the actions you add here will run.

Status: Published

Application: Global

Name	Application	State	Installed from	Created	Created by	Parent	Batch Base
Default	Global	In progress		2026-06-13 02:55:17	system	(empty)	(empty)
Project Update Set	Global	Complete		2026-09-28 06:53:47	admin	(empty)	(empty)

Related Links
[Merge Update Sets](#)

1 to 2 of 2

8. ADVANTAGES & DISADVANTAGES

Advantages

- Automatically classifies IT tickets.
- Reduces manual work for IT Helpdesk staff.
- Provides consistent Category and Subcategory assignment.
- Supports dependent Category–Subcategory choices.
- Sends automatic email notifications to callers.
- Stores ticket information in a structured format.
- Easy to maintain and extend with additional issue types.

Disadvantages

- Classification depends on the issue description and configured conditions.
- Incorrect or unclear descriptions may affect classification.
- Requires ServiceNow configuration and access.
- Changes to categories or classification rules require administrative updates.
- Email notification depends on the ServiceNow notification configuration.

9. CONCLUSION

The Automated School IT Ticket Classification and Notification System successfully addresses the problem of manual ticket classification in the school IT Helpdesk. The system automatically identifies common IT issues, assigns the appropriate Category and Subcategory, updates the ticket, and sends an email notification to the Caller.

The implemented solution improves consistency, reduces repetitive manual effort, and provides a structured approach for handling IT support requests.

10. FUTURE SCOPE

The system can be further enhanced by:

- Adding more IT issue categories and subcategories.
- Using AI or machine learning for advanced ticket classification.
- Supporting more complex issue descriptions.
- Adding dashboards for monitoring ticket statistics.
- Integrating automated assignment to specific IT support teams.
- Adding priority and urgency prediction.
- Providing SLA tracking and escalation.
- Integrating chatbot or virtual assistant support.
- Adding analytics to identify frequently occurring IT problems.
- Extending the system for mobile-based ticket management.

GITHUB LINK: <https://github.com/bharsha525/Auto-Ticket-Classification-Using-Flow-Designer.git>

DEMO LINK: